

MUHAMMAD SHAHZAD ALI

Product Engineer · full stack, ships end to end

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I build products on my own and put them in front of users: frontend, backend, deploy, and the glue in between. TypeScript and Python daily. I have not shipped SvelteKit yet, and I rebuilt my entire site in a day this week, so the ramp is short.

EXPERIENCE

Software Engineer [Grow_in \(Kairo / Fable\)](#)

Mar 2026 – Jul 2026

- Architected Kairo, an AI video generation platform (React + TypeScript Tauri desktop app on a FastAPI backend) that turns a prompt into a cut-ready film, reducing manual production effort by 80%.
- Built the pipeline in three stages: a Writer producing 10-scene scripts through OmniRoute LLM routing, a TTS module aligning narration with edge_tts word-boundary timestamps, and an Aligner checking each scene against a 2.25 words/sec baseline, then generated the edit decision list straight from the JSON scenes.
- Shipped Fable, a Shopify app with three custom storefront components, and containerised the stack with CI/CD so staging and production each deploy with one command.

Software Engineer [Mailtrail, Inc. \(Contract\)](#)

Dec 2024 – Jan 2026

- Built an AI assessment product: an organization adds its employees and students, an LLM writes the test, it goes out from the organization's own email domain, and graded answers come back into React dashboards.
- Ran dispatch and evaluation as async jobs on Redis-backed queues. Deployed to AWS and cut monthly cost by roughly 30% by rewriting the slowest database queries.

Software Engineer [Fiverr \(Freelance\)](#)

Feb 2023 – Mar 2025

- 5+ web apps delivered for clients in the USA and Europe at a 100% completion rate and a 5-star rating, including a console resale marketplace, a movers site and an Amazon affiliate store. Took one legacy Node.js and MongoDB backend from 3.2s to 400ms.

BUILT AND SHIPPED

Cetus, multi-provider AI media platform [cetus-one.vercel.app](#)

Live in production

- Built and shipped the whole product solo. One composer where a user types a prompt and gets stills, video or voice back, with one internal API in front of Flow, Qwen and edge-tts so the provider choice never reaches the interface.
- Frontend in Next.js App Router with Tailwind and TanStack Query. Every running job streams progress over a single Server-Sent Events connection instead of a poll per job, which is what made the composer feel live.
- Backend in FastAPI on async SQLAlchemy 2.0 with Postgres/Supabase, Alembic migrations and an in-process asyncio worker pool. Supabase Auth with Google OAuth, credit-based billing, usage analytics and an admin console.
- Runs across Vercel, GCP and Supabase. Built end to end with Claude Code: the repo carries a CLAUDE.md, custom skills, hooks, review subagents and MCP servers.

Personal site [muhammadshahzadali.netlify.app](#)

Built in a day

- Designed and shipped a personal site from an empty repo in one working day, working through Claude Code with parallel subagents: one measured a reference site with a headless browser, others built the object primitives, the content layer and each route at the same time, and a verification pass screenshotted all 27 pages at two widths and fixed what it found.
- Every image on it is generated from code and seeded per slug, the diagrams are hand-inlined SVG, and the commit history is public.

RESEARCH

- First author on "Adaptive Hybrid Quantum-Classical Framework with Byzantine-Robust Federated Learning and Noise-Aware QAOA", accepted at ICATCICT 2026 in Dubai (25 to 27 November 2026), to appear in AIP Conference Proceedings, Scopus and Web of Science indexed. Cut an operating-room scheduling optimality gap from 5.1% to 0.9% by choosing circuit depth from live hardware calibration.

SKILLS

- Frontend:** TypeScript, React, Next.js (App Router), Tailwind, TanStack Query, Server-Sent Events, Tauri. Svelte and SvelteKit: reading the docs this week, not yet shipped.
- Backend and data:** Python, FastAPI, Node.js, async SQLAlchemy 2.0, PostgreSQL, Supabase, Redis, MongoDB, Alembic.
- AI:** Claude Code and agent workflows, MCP servers, LLM routing, RAG, Chroma, PyTorch, Qiskit, PennyLane.
- Infra:** Vercel, GCP, AWS (EC2, S3), Docker, CI/CD, Playwright.

EDUCATION

BS in Artificial Intelligence [University of Management and Technology, Lahore](#)